

Name _____ Date _____ Period _____

Modeling with a Parent Function: the Open Box

Algebra II Honors | Unit 1 | Day 9 | TEKS A2.2.A, A2.7.I

Objective: I can model a situation with a cubic function, justify its contextual domain, locate a maximum numerically, and generalize the model to any rectangular sheet.

The Problem

A flat sheet of cardboard is **20 in. by 10 in.** A square of side x is cut from each corner and the flaps are folded up to form an open-top box.

Part 1 — Build and Expand

Each cut removes x from BOTH ends of a side. Write each dimension, then the volume.

height _____ length _____ width _____

$V(x) =$ _____

Now expand to standard form. Show the multiplication.

$V(x) =$ _____ Degree _____ Family _____

What are the three zeros of the expanded form, and which are physically meaningful?

Zeros at $x =$ _____ meaningful: _____

Part 2 — Justify the Domain

Why $x > 0$? _____

Why $x < 5$ and not $x < 10$? _____

Contextual domain _____ Open or closed, and why? _____

Part 3 — Locate the Maximum

Compute $V(x)$. Round to the nearest hundredth where needed.

x	1	2	2.1	2.11	2.2	3	4
$V(x)$							

Average rate of change on $[1, 2]$: _____ on $[2, 3]$: _____

The sign flips between those intervals. What does that tell you, and why is it stronger evidence than just reading the largest table value?

The volume stops rising and starts falling, so a maximum lies _____.

Maximum $\approx$ _____ at $x \approx$ _____ Box $\approx$ _____

Range on this domain: _____ Why is 0 excluded but 192.45 included?

Part 4 — Generalize

Now let the sheet be **L by W** with W the shorter side, and squares of side x cut from the corners.

$V(x) =$ _____ Contextual domain _____

Explain why the upper bound depends on W and never on L:

The shorter side reaches zero first, so it _____.

Apply it: a 16 by 8 sheet gives domain _____ a 30 by 30 sheet gives domain _____

Part 5 — Going Further

1. For the 16 by 8 sheet, compute $V(1)$ and $V(2)$. Show the arithmetic.

$V(1) =$ _____ $V(2) =$ _____

2. For a square sheet s by s , write $V(x)$ and its domain. $V(x) =$ _____ domain _____

3. Does doubling both sheet dimensions double the maximum volume? Predict, then justify.

No — _____

4. A sheet is 18 by 6. Find the domain, then compute V at the midpoint of that domain.

Domain _____ midpoint $x =$ _____ $V(1.5) =$ _____

5. Is the maximum always at the midpoint of the domain? Test against the 20 by 10 sheet.

No — _____

Exit Ticket

A 12 in. by 12 in. sheet has squares of side x cut from each corner.

$V(x) =$ _____ Domain _____ $V(2) =$ _____

Explain the upper bound in one sentence: _____